INCH-POUND
MIL-DTL-24643/66
7 November 2007

DETAIL SPECIFICATION SHEET

CABLE, ELECTRICAL, 1000 VOLTS, TYPE LS2OW

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-24643.

REQUIREMENTS:

Qualification Required.

Construction (Watertight with Circuit Integrity)

Sizes 3 to 200 Inclusive, Thermoset Insulation

First	-	Copper conductor, coated or uncoated (see Table I for size).
Second	-	Thermoset insulation, extruded or taped (see Table I for minimum average wall).
Third	-	Optional glass braid.
Fourth	-	Optional covering, unless required (see 3.3.5 of MIL-DTL-24643).
Fifth	-	Two conductors shall be cabled together to form a firm well-rounded assembly with a lay not less than 8 nor greater than 16 times the cabled diameter. Standard identification code applied by Method 1, 3, or 5. Fillers may be used as necessary to form a firm well rounded assembly.
Sixth	-	Aluminum foil/polyester tape/aluminum foil. 0.002-inch diameter with 45 percent overlap.
Seventh	-	Braided shield of tin-coated copper. AWG and coverage as necessary to meet shield performance requirements.
Eighth	-	An optional binder.
Ninth	-	Cross-linked polyolefin jacket (see Table I for wall thickness).

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TABLE I. Details.

Military Part No. M24643/66 (TYPE)	Conductors		Insulation Wall Thickness Min. Avg. (inches)	Jacket Wall Thickness Min. Avg. (inches)	Overall Diameter (inches)		Conductor Resistance per 1000 Feet at 25 °C (ohms)
	Size (AWG)	Min. No. of Strands			Min.	Max.	
-01UO (LS2OW-3)	16	7	0.018	0.027	0.345	0.370	4.3
-02UO (LS2OW-4)	14	7	0.018	0.028	0.366	0.392	2.68
-03UO (LS2OW-9)	10	19	0.018	0.036	0.466	0.499	1.08
-04UO (LS2OW-14)	9	37	0.025	0.050	0.639	0.686	0.859
-05UO (LS2OW-23)	7	61	0.025	0.058	0.743	0.797	0.543
-06UO (LS2OW-50)	3	61	0.035	0.069	0.894	0.959	0.21
-07UO (LS2OW-75)	1	91	0.035	0.080	1.033	1.108	0.134
-08UO (LS2OW-100)	0	91	0.035	0.089	1.149	1.232	0.106
-09UO (LS2OW-200)	0000	127	0.050	0.100	1.519	1.630	0.053

EXAMINATION AND TESTS:

Basic Electricals:

Conductor resistance (ohms/1000 feet at 25 °C, max.)	see Table I
Voltage withstand (volts, root mean square, min.)	
Conductor to conductor (sizes 3 – 9)	3000
Conductor to conductor (sizes 14 – 200)	5000
Conductor to shield (sizes 3 – 9)	1500
Conductor to shield (sizes 14 – 200)	2500
Insulation resistance (megohms/1000 feet, min.)	
Sizes 3 – 23	500
Sizes 50 – 200	250
Conductor and shield continuity	No Failure
Jacket flaws	No Failure

Group A:

Visual and dimensional	No Failure
Watertightness (see MIL-DTL-24643 for limits of water leakage)	No Failure
Crack resistance (applicable to constructions using glass braids with coverings)	No Damage

Group B:

Thermoset proof test (percent, max.)

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Insulation (extruded insulations only)	50
Jacket (when tested at 200 °C)	50
Drip (95±1 °C)	Zero
Cold bending (cable)	No Damage
Gas flame (1 hour)	No Failure
Physicals (unaged)	
Insulation (extruded insulations only)	
Tensile strength (lb/in², min.)	1300
Elongation (percent, min.)	150
Jacket (cable)	
Tensile strength (lb/in², min.)	1300
Elongation (percent, min.)	160
Tear (pounds per inch thickness, min.)	35
Group C:	
Physicals (aged)	
Jacket (cable)	
Air oven	
Tensile strength (percent of unaged, min.)	60
Elongation (percent of unaged, min.)	60
Shrinkage	No Failure
Heat distortion (percent of unaged, max.)	30
Permanence of printing (conductor – Method 1 only) (cycles, min.)	25
Permanence of printing (jacket) (cycles, min.)	125
Cable sealant removability	No Failure
Surface transfer impedance	
Milliohms per meter, max.	700
EMP, response (db, min.)	60
Group D:	
Flame propagation (cable)	No Failure
Qualification Inspection:	
Qualification inspection shall include basic electricals, groups A, B, C, and D, plus the following:	
Gas flame (3 hours)	No Failure

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Aging and compatibility (cable) (125±5 °C)	No Failure
Abrasion resistance	
Jacket-75 scraps minimum	No Damage
Insulation-extruded insulation only-250 scraps minimum	No Damage
Acid gas equivalent (percent, max.)	
Jacket	2
Fillers	2
Insulation	18
Halogen content (percent, max.)	
Jacket	0.2
Fillers	0.2
Insulation	0.2
Immersion (jacket)	
Tensile strength (percent of unaged, min.)	50
Elongation (percent of unaged, min.)	50
Smoke index, max.	
Jacket	25
Fillers	45
Insulation	35
Toxicity index, max.	
Jacket	5
Fillers	5
Insulation	1.5
Durometer (jacket) – Type A (hardness, min.)	80
Weathering (jacket)	No Failure

UNIT ORDERING LENGTHS: 500 feet (nominal).

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Custodians:
Army – MI
Navy – SH

Preparing Activity:
Navy – SH
(Project 6145-2007-012)

Review Activities:
Army – AR, AV, CR
Navy – CG, EC
DLA – CC

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at http://assist.daps.dla.mil.

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